

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • Stamp 1G — Full-Time Work Eligible

PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Designed and implemented a **Python**-based data ingestion pipeline on **AWS Lambda** to process approximately **3,500 events/sec**, replacing a less efficient batch job.
- Developed and deployed a **FastAPI** service for ML model inference, achieving a p95 latency of **180ms** and integrating with an internal feature store.
- Engineered a **Python** script to automate the generation of daily operational reports, reducing manual effort by **8 hours per week** for the financial operations team.
- Contributed to the development of the **RAIMS.chat** agent interface, implementing features for streaming responses and user feedback mechanisms.
- Built a **Python** data extraction and transformation process for sourcing legally usable knowledge from PDFs and HTML documents, preparing content for a vector database.
- Implemented advanced reasoning frameworks such as **Chain-of-Thought** and **ReAct** within prompt engineering experiments to enhance LLM response accuracy.
- Containerized **Python** services using **Docker** and deployed them to an **AWS EKS** cluster, configuring basic resource limits for predictable deployments.

SKILLS

Languages – Python, Java, SQL, R

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch, EKS), Docker, Kubernetes, Jenkins, Git

AI/ML – LLMs, RAG, Embeddings, Vector Databases, Prompt Engineering, LangChain, scikit-learn, Machine Learning Fundamentals

Data Analytics & BI – Power BI, Tableau, DAX, Power Query, Data Modelling, Data Warehousing, Pandas, NumPy

Backend Frameworks – Spring Boot, FastAPI, Flask

Databases – PostgreSQL, MySQL, Oracle

Tools & Methodologies – Jira, Confluence, Agile, Scrum, ISO/IEC 42001, EU AI Act

PROJECTS

RAIMS.chat Agent Interface Development

- Developed the chat UI for RAIMS.chat, incorporating persona selection, streaming output, and citation display to enhance user experience.
- Implemented agent orchestration logic using functions and RAG, enabling the system to generate explainable and traceable answers.
- Integrated logging mechanisms to track prompt metadata, latency, and risk flags, supporting governance and audit requirements.

Responsible AI Knowledge Graph Ingestion

- Built a **Python** pipeline to ingest and process diverse data sources, including PDFs and HTML documents, for the RAIMS knowledge base.
- Developed processes for chunking, tagging, and storing content in a **vector database**, forming the trusted data layer for RAG applications.
- Designed gold-standard test sets and verified answers to evaluate grounding precision and citation fidelity within the AI system.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)